Diamante

Networking & Storage Prototype using P2P & Ledger Modules

TEAM NAME: TECHSENATE

About Project

- Developed a prototype for the Diamante L1 blockchain stack.
- Networking (P2P): Established a decentralized peer-to-peer network for node communication.
- Storage (Ledger/State Machine): Managed immutable transaction records and maintained accurate state.
- Implemented real-time transaction processing and data persistence.
- Created a visual dashboard to monitor node interactions and ledger updates.

PLAN

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Backend

Networking + Storage

Backbone for Communication + state consistency in Blockchain

VS Code

GitHub

Ext.

Golang (P2P)

1) node.go
2) ledger.go
3) dashboard.go

P2P → Golang.

Ledger

Database Module (Table format)

↓ MongoDB.

LevelDB / Rocks DB

↓

easier to setup with Go.

Frontend (minimal).

⟹ DASHBOARD (Web page)

Main goal: node communication flow.

HTML, CSS, Javascript

Node 1

msg sent

→

Node 2

msg recieved

TRANSACTION

↓

Gossip Rate + Broadcast

Table in Ledger

idx	msg	status.

Scalability. ??

What abt broadcast not just "2"?

Output:

Live Transaction Table + Node communication.

Node 1 $\xrightarrow{\text{send msg}}$ Gossip Layer (Golang P2P) $\xrightarrow{\text{recieve msg}}$ Node 2

Fault Tolerance

N2 is down.

↓

Retry Mechanism

Success / failed.

Log event

↓

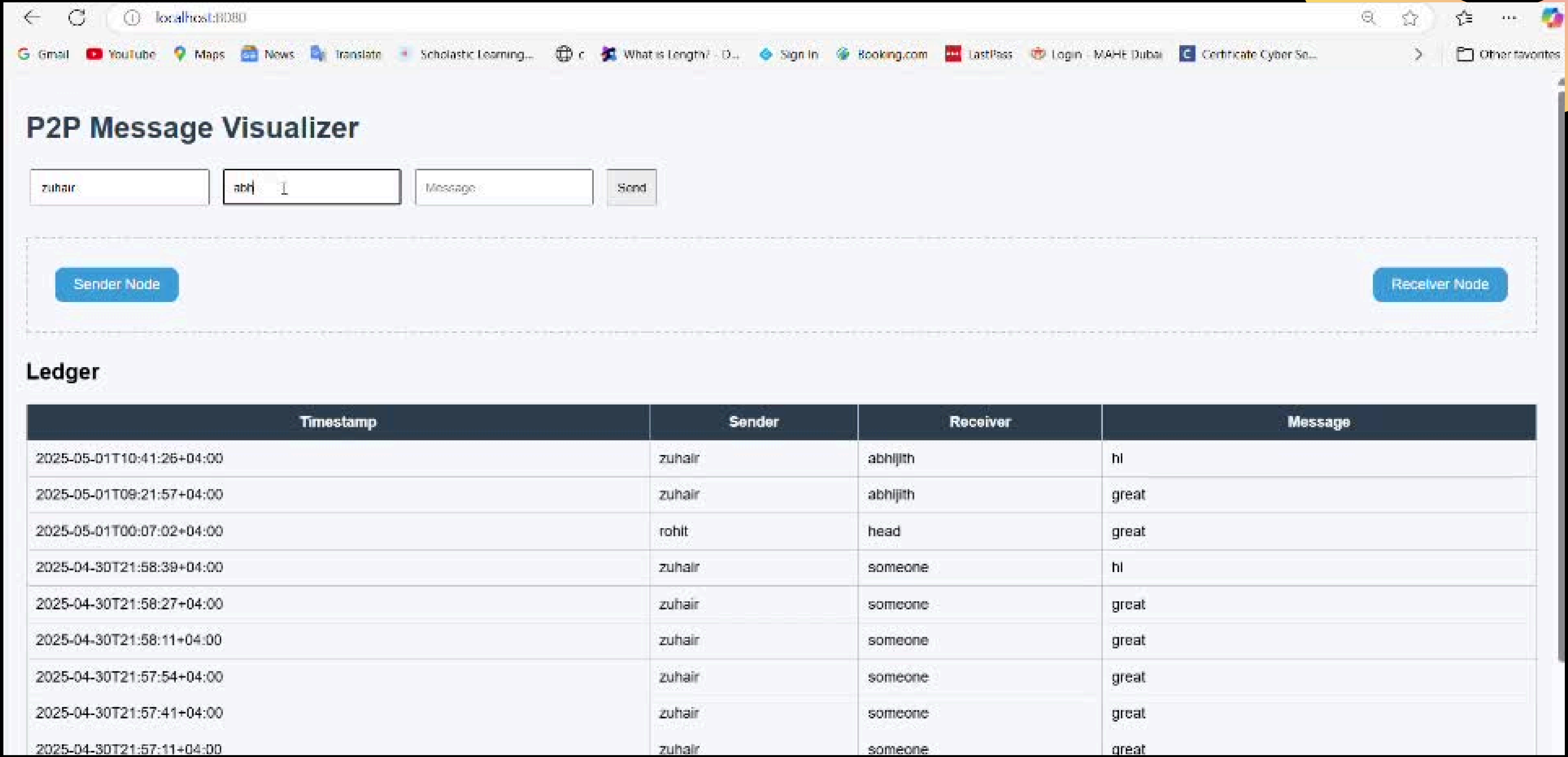
Ledger Table → DB (LevelDB)

Dashboard: Minimal.

↓

WebSocket / EventSource

Demo Video


localhost:8080
Gmail
YouTube
Maps
News
Translate
Scholastic Learning...
What is Length? - D...
Sign In
Booking.com
LastPass
Login - MAHE Dubai
Certificate Cyber Se...
Other favorites
P2P Message Visualizer
zuhair
abh
Message
Send
Sender Node
Receiver Node
Ledger
Timestamp
Sender
Receiver
Message
2025-05-01T10:41:26+04:00 zuhair abhijith hi
2025-05-01T09:21:57+04:00 zuhair abhijith great
2025-05-01T00:07:02+04:00 rohit head great
2025-04-30T21:58:39+04:00 zuhair someone hi
2025-04-30T21:58:27+04:00 zuhair someone great
2025-04-30T21:58:11+04:00 zuhair someone great
2025-04-30T21:57:54+04:00 zuhair someone great
2025-04-30T21:57:41+04:00 zuhair someone great
2025-04-30T21:57:11+04:00 zuhair someone great

System Components:

- Node 1: Sends transactions/messages.
- Node 2: Receives and stores messages in LevelDB.
- Server: Exposes stored data via REST API.
- Dashboard: Displays live updates of node messages and ledger entries.

Technologies Used:

- GoLang for networking and database operations.
- LevelDB for lightweight key-value storage.
- HTML/CSS/JS for the frontend interface.

Real-world Application

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Money Transfer

Recipient Name / ID

Abhi

Amount in USD

400

Send Money

Node 1 (USER)

Receiver Node

Ledger

From	To	Amount
USER	Abhi	$200.00
USER	Zuhair	$200.00
USER	2345674	$2345.00

Key Features & Outcomes

- Simulated peer-to-peer node communication.
- Real-time gossip message transmission.
- Persistent storage using LevelDB.
- Integrated API and frontend for seamless interaction.
- Modular design separating nodes, server, and dashboard.
- Outcome: Functional minimum viable product demonstrating node communication and ledger operations, providing a foundation for further enhancements like consensus mechanisms and scalability.

Industry Impact

1. Faster & Reliable Blockchain Nodes

What it means: Blockchain networks consist of many nodes (computers) that store and verify transactions.
Impact: Improving their speed and reliability means transactions are processed quicker and the network is more stable.
Why it matters: This is essential for making blockchain systems scalable and user-friendly (e.g., faster payments).

2. Stronger Network Security

DoS Protection: Prevents "Denial of Service" attacks, which try to flood the network with fake traffic and shut it down.
Gossip Protocol: A method nodes use to spread information quickly and efficiently, like how rumors spread in a crowd.
Impact: Together, they make the network more secure and ensure all nodes stay updated reliably.

3. Smarter Ledger Design (using LwDAP)

LwDAP: A framework or method that helps in analyzing and choosing the best structure for the blockchain ledger.
Ledger: A digital record of all transactions in the blockchain.
Impact: Leads to better organization, performance, and adaptability of the blockchain system.

4. Trends & Evolution Insight

What it does: Monitors and analyzes changes in the blockchain industry.
Impact: Helps companies and developers stay updated with new technologies, upgrades, or standards.
Why it matters: Staying ahead ensures competitiveness and innovation.

5. Testing & Simulation

Best-of-breed testing: Using top-quality tools to simulate real-world blockchain behavior.
Impact: Lets developers test their networks safely before launch, finding bugs or weaknesses early.
Why it matters: Prevents costly failures and improves user trust.

Web3 is a decentralized version of the internet, where users have control over their data, identities, and assets using blockchain technology.

1. Decentralization- No central authority (like Facebook or Google); instead, apps run on blockchains or peer-to-peer networks.
2. Ownership-Users truly own their data, digital items (like NFTs), and money (like crypto wallets).

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Thank You

Thanks for this amazing opportunity!
Diamante V2 Hackathon